# CTF WRITEUP

Havoc C2 Framework — Network Forensics

TryHackMe — Packet Analysis / Wireshark – Achira Thilakarathne

## 1. Overview

The hex value deadbeef is present in the captured packets, identifying the Command & Control framework as Havoc C2 — a post-exploitation framework tied to the TrickBot/Ryuk ransomware ecosystem.

**How to locate the marker in Wireshark:**

Edit → Find Packet → Search by Hex Value → enter: deadbeef

## 2. Packet 182 — Initial Beacon (Client → Server)

Packet 182 contains the first occurrence of the deadbeef magic bytes. This is a POST request from 10.0.2.38 to 10.0.2.37, confirming 10.0.2.38 is the client beaconing to the C2 server.

**Action: Copy Hex Stream from Packet 182**

### Hex Stream Breakdown

```
00000113
deadbeef
0e9f (skip)
b7d8
0000006300000000 (skip)
946cf2f65ac2d2b868328a18dedcc296cc40fa28fab41a0c34dcc010984410ca  ← AES Key
8cd00c3e349290565aaa5a8c3aacd430  ← IV Key
```

**Data field (encrypted beacon payload — excerpt):**

```
340cd2bc567802a30e1c2722bb2135dbfe9b76cc0f486e975aaf1f21a6ad13f9
ef282131470a0191c974b4f5c2369005227811dcfb9708192e358fe9dfbf4214
760291da15237a0f2e5db5b1e718d48ef01af5e724767c95a1b5c136cc744d75
edcb1747847d1b6a22df5a23531442b73329c2b2ae46e1ab2506f548bd1b7068
... (full stream in Wireshark)
```

**HTTP headers (hex-decoded from same packet):**

```
POST / HTTP/1.1
Cache-Control: no-cache
Connection: Keep-Alive
Pragma: no-cache
Content-Type: */*
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/96.0.4664.110 Safari/537.36
Content-Length: 5781
Host: 10.0.2.37
```

## 3. Packet 239 — SID Extraction

The hex stream from Packet 239 was decrypted using the AES key and IV extracted from Packet 182. The decrypted output reveals the victim's Windows Security Identifier (SID).

**Note: For client→server packets, copy Hex Stream. For server→client packets, copy the Data field.**

**Hex stream (excerpt — full stream in Wireshark):**

```
0e9fb7d80000000100000003a93653eecbbd285595568438269
0ca5bea610e8236938e0298f5301e78d13978941465456071c9d
c86eca9abd4af87d497811d489f96d643f64c5b881f909556e47
e5b2361214505a3bb8eeea61f4cbd02b9afa4a4f1f91c8bbaf78
... (truncated)
```

**Extracted SID:**

```
S-1-5-21-679395392-3966376528-1349639417-1103
```

## 4. Server Network Information

The hex stream from the server reply (copy Data field, not Hex Stream) was decrypted to reveal the server's full network configuration.

| Field | Value |
|---|---|
| **Adapter** | `Ethernet` |
| **DNS Suffix** | `home` |
| **Link-local IPv6** | `fe80::e134:1b0c:c8d5:3020%6` |
| **IPv4 Address** | `10.0.2.38` |
| **Subnet Mask** | `255.255.255.0` |
| **Default Gateway** | `10.0.2.1` |

## 5. Packet 314 — System Information

Decrypted output from Packet 314 reveals the full systeminfo output of the compromised host.

| Field | Value |
|---|---|
| **Host Name** | `WIN-9H86M71MBE9` |
| **OS Name** | `Microsoft Windows Server 2019 Standard Evaluation` |
| **OS Version** | `10.0.17763 N/A Build 17763` |
| **OS Configuration** | `Primary Domain Controller` |
| **System Manufacturer** | `innotek GmbH` |
| **System Model** | `VirtualBox` |
| **System Type** | `x64-based PC` |
| **Processor** | `Intel64 Family 6 Model 158 Stepping 13 GenuineIntel ~3600 MHz` |

| Field | Value |
|---|---|
| Windows Directory | C:\Windows |
| System Locale | en-us; English (United States) |
| Time Zone | (UTC-08:00) Pacific Time (US & Canada) |
| Total Physical Memory | 8,192 MB |
| Domain | clientserver.thm |
| Logon Server | \\WIN-9H86M71MBE9 |
| Hotfixes Installed | KB5020627, KB5019966, KB5020374 |
| NIC | Intel(R) PRO/1000 MT Desktop Adapter |
| DHCP Server | 10.0.2.3 |

# 6. Flags & Credentials Found

## Flag 1 — Packet 341 (Printed Output)

```
THM{HavOc_C2_DeCRypTing_is_Fun_Fun_FUN}
```

## Flag 2 — Packet 610 (File Exfiltration)

**Important file path (Packet 602):**

```
C:\Users\paco\Desktop\Files\clients.csv
```

```
THM{I_Can_SEE_ThE_fiL3_YoU_ToOk}
```

## Attacker-Created Credentials

| Username | Password |
|---|---|
| administrato | WfD3hz3AXZ4n |